

Interface Control Document — Module Requirements

Document Reference: SEC-IF-007

1 Introduction

This document specifies the module-level requirements for the Interface Control Module (ICM). Requirements are captured in tabular form with the acceptance criterion and rationale for each requirement.

2 Module Requirements

The following table defines all module requirements. Each row constitutes a single traceable requirement.

Module Requirement ID	Source of Requirement	Module Requirement Acceptance Criterion	Rationale/ Supporting information
MR-001	SYS-SPEC-001 §3.1	The module shall accept digital input signals in the range 0 V to 3.3 V with a maximum input impedance of 10 kΩ.	Compatibility with existing 3.3 V logic families in the platform.
MR-002	SYS-SPEC-001 §3.2	The module shall provide galvanic isolation of at least 1500 V AC RMS between input and output circuits.	Safety requirement to protect downstream logic from power-line transients.
MR-003	SYS-SPEC-002 §4.1	The module shall operate within the temperature range -40 °C to +85 °C without performance degradation.	Deployed in outdoor enclosures subject to wide ambient temperature variation.
MR-004	SYS-SPEC-002 §4.2	The module shall maintain its rated accuracy after 1000 thermal cycles from -40 °C to +85 °C.	Long-term reliability requirement for 10-year service life.
MR-005	SYS-SPEC-003 §2.1	The module shall communicate over CAN bus at 500 kbit/s in accordance with ISO 11898-2.	CAN is the standard bus for this vehicle platform; 500 kbit/s is mandated by the platform integration specification.
MR-006	SYS-SPEC-003 §2.2	The module shall respond to a CAN wake-up frame within 5 ms of receipt.	Response latency budget allocated by the system power-management specification.
MR-007	SYS-SPEC-004 §5.1	The module firmware shall be updateable in the field via the CAN bootloader interface without removal from the enclosure.	Reduces field maintenance cost; a physical removal risk was identified in the maintenance hazard analysis.
MR-008	SYS-SPEC-004 §5.2	The module shall report a CRC failure and revert to the previous firmware image if a firmware update checksum does not match.	Prevents bricking of deployed units due to corrupted update packages.
MR-009	SYS-SPEC-005 §6.1	The module shall draw no more than 150 mA from the 5 V supply rail under maximum load conditions.	Power budget constraint allocated in the system power distribution analysis.
MR-010	SYS-SPEC-005 §6.2	The module shall tolerate supply voltage variations from 4.5 V to 5.5 V without entering an error state.	Accounts for voltage drop across connectors and wiring harness under worst-case resistance.
MR-011	SYS-SPEC-006 §7.1	The module shall conform to IPC-A-610 Class 3 workmanship standards for all solder joints and component placements.	Class 3 is required for safety-critical automotive applications per the quality assurance plan.
MR-012	SYS-SPEC-006 §7.2	The module shall pass EMC testing to CISPR 25 Class 5 limits for radiated and conducted emissions.	Regulatory and customer requirement for use in passenger vehicles.

3 Verification Summary

Each requirement in Section 2 shall be verified by test in accordance with the Module Verification Plan (MVP-ICM-001). Traceability between requirements and verification activities is maintained in the Requirements Traceability Matrix.